LAB Logbook

Lab 1

```
#SID is 2520731, so the last two digits are 31.

#1)Create a vector using np.arange.
a = np.arange(31)
print(a)

#2)Change a to a 2-D array with 1 row
a1 = a.reshape(1, 31)
print(a1)

#3)Save it in another array. Print the array.
a2 = a1
print(a2)

#4)Check the shape attribute value.
print(a2.shape)
```

```
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
 24 25 26 27 28 29 30]
[[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
  24 25 26 27 28 29 30]]
[[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
  24 25 26 27 28 29 30]]
(1, 31)
```

Lab 2

```python
# 1. Determine n from the last digit of the SID=2520731
n = 1

# 2. Group by relationship and hours-per-week
Group_by_relationship_hours = data.groupby(["relationship", "hours-per-week"])
Group_by_relationship_hours.size()

# 3. Reduce hours-per-week by n using a function
def reduce_hours(x):
    return x - n

data["hours-per-week-reduced"] = data["hours-per-week"].apply(reduce_hours)

data

# 4. Group by relationship and reduced hours-per-week
Group_by_relationship_reduced_hours = data.groupby(
    ["relationship", "hours-per-week-reduced"]
)

Group_by_relationship_reduced_hours.size()
```

```
                                        0
relationship  hours-per-week-reduced
Husband       12.0                      1
              39.0                      2
              44.0                      1
              79.0                      1
Not-in-family 15.0                      1
              39.0                      2
              49.0                      2
Own-child     29.0                      1
Wife          39.0                      2
dtype: int64
```

Lab 3

jupyter Week-3-Pandas-and-Seaborn_ML_in-Finance_Final-1 (1) Last Checkpoint: 3 minutes ago

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Code JupyterLab Python 3 (ipykernel)

[46]:
```
# Bicolour features interaction diagram
# SID = 2520731
# Last digit = 1  → Account length
# Second-to-last digit = 3 → Total day charge

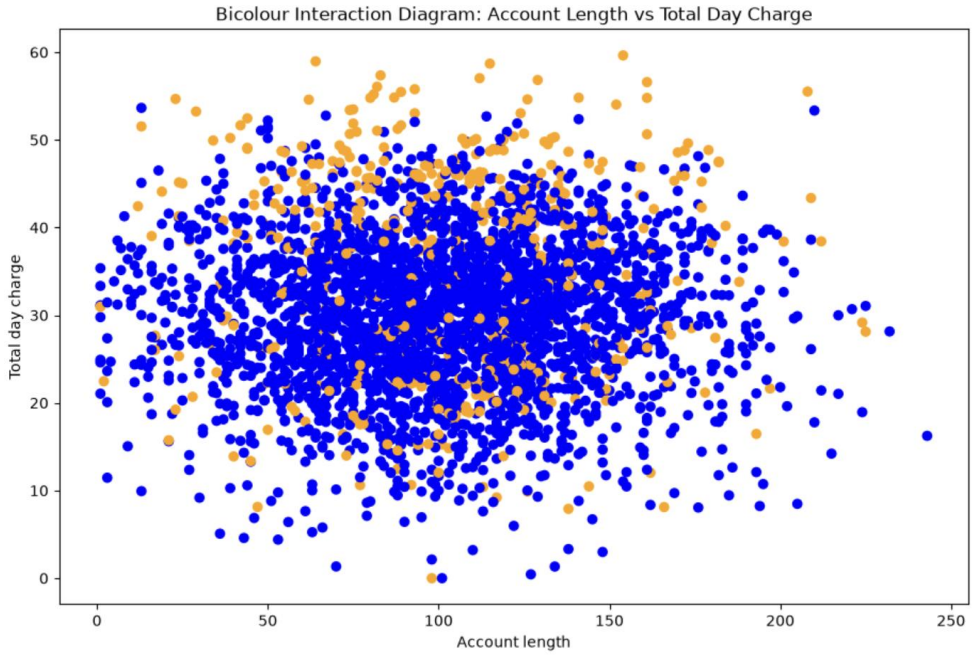
Clr = data['Churn'].map({False: 'blue', True: 'orange'})

fig = plt.figure(figsize=(11,7))

plt.scatter(data['Account length'],
            data['Total day charge'],
            color=Clr)

plt.xlabel('Account length')
plt.ylabel('Total day charge')
plt.title('Bicolour Interaction Diagram: Account Length vs Total Day Charge')

plt.show()
```



Lab 4

MLP architecture:

[70]:
```
# Create an MLP with two hidden layers based on my SID
# SID: 2520731
# First hidden layer: 731 neurons
# Second hidden layer: 366 neurons

model = keras.Sequential([
    keras.Input(shape=(X_train.shape[1],)),
    keras.layers.Dense(731, activation='relu'),
    keras.layers.Dense(366, activation='relu'),
    keras.layers.Dense(1)
])

# Display the model architecture
model.summary()
```

Model: "sequential_4"

Layer (type)	Output Shape	Param #
dense_17 (Dense)	(None, 731)	366,231
dense_18 (Dense)	(None, 366)	267,912
dense_19 (Dense)	(None, 1)	367

Total params: 634,510 (2.42 MB)

Trainable params: 634,510 (2.42 MB)

Non-trainable params: 0 (0.00 B)

MLP training code and process:

[73]:
```
#Train the MLP for 10 epochs

history = model.fit(
    X_train,
    y_train,
    batch_size=10,
    epochs=10,
    validation_split=0.2,
    verbose=1
)
```

```
Epoch 1/10
2640/2640 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 0.0070 - mae: 0.0191 - val_loss: 0.0198 - val_mae: 0.1237
Epoch 2/10
2640/2640 ━━━━━━━━━━━━━━━━━━━━ 5s 2ms/step - loss: 2.8603e-04 - mae: 0.0128 - val_loss: 0.0086 - val_mae: 0.0819
Epoch 3/10
2640/2640 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 1.7039e-04 - mae: 0.0097 - val_loss: 0.0040 - val_mae: 0.0568
Epoch 4/10
2640/2640 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 1.1358e-04 - mae: 0.0082 - val_loss: 0.0037 - val_mae: 0.0555
Epoch 5/10
2640/2640 ━━━━━━━━━━━━━━━━━━━━ 7s 3ms/step - loss: 9.3823e-05 - mae: 0.0075 - val_loss: 0.0020 - val_mae: 0.0402
Epoch 6/10
2640/2640 ━━━━━━━━━━━━━━━━━━━━ 7s 2ms/step - loss: 7.6295e-05 - mae: 0.0065 - val_loss: 0.0010 - val_mae: 0.0279
Epoch 7/10
2640/2640 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.7249e-05 - mae: 0.0062 - val_loss: 0.0013 - val_mae: 0.0320
Epoch 8/10
2640/2640 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.0549e-05 - mae: 0.0059 - val_loss: 0.0012 - val_mae: 0.0313
Epoch 9/10
2640/2640 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.3413e-05 - mae: 0.0060 - val_loss: 5.9955e-04 - val_mae: 0.0209
Epoch 10/10
2640/2640 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 5.6988e-05 - mae: 0.0057 - val_loss: 2.5761e-04 - val_mae: 0.0127
```

Resulting MAE on the test dataset:

[74]:
```
#Evaluate the model on the test dataset

mse, mae = model.evaluate(X_test, y_test, verbose=0)

print("Mean absolute error: %.5f" % mae)
print("Mean squared error: %.5f" % mse)
```

```
Mean absolute error: 0.03683
Mean squared error: 0.00174
```

[75]:
```
#Compare your MAE with the practical model

practical_mae = 0.02932

print("Practical model MAE:", practical_mae)
print("My MLP test MAE:", round(mae, 5))

if mae < practical_mae:
    print("My MLP achieved a lower test MAE than the practical model.")
elif mae > practical_mae:
    print("My MLP achieved a higher test MAE than the practical model.")
else:
    print("Both models achieved the same test MAE.")
```

```
Practical model MAE: 0.02932
My MLP test MAE: 0.03683
My MLP achieved a higher test MAE than the practical model.
```

Lab 5

[39]:
```
mae_practical = mae      # keep the practical session's test MAE for comparison

model_lab = keras.Sequential([

    keras.layers.Input(shape=(50, 5)),

    keras.layers.Conv1D(50, 5, padding='same', activation=tf.nn.relu, kernel_initializer="normal"),

    keras.layers.MaxPooling1D(7),

    keras.layers.Conv1D(100, 5, padding='same', activation=tf.nn.relu, kernel_initializer="normal"),

    keras.layers.GlobalMaxPooling1D(),

    keras.layers.Dense(25, activation=tf.nn.relu, kernel_initializer="normal"),

    keras.layers.Dense(2)
])

model_lab.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv1d_2 (Conv1D)	(None, 50, 50)	1,300
max_pooling1d_1 (MaxPooling1D)	(None, 7, 50)	0
conv1d_3 (Conv1D)	(None, 7, 100)	25,100
global_max_pooling1d_1 (GlobalMaxPooling1D)	(None, 100)	0
dense_2 (Dense)	(None, 25)	2,525
dense_3 (Dense)	(None, 2)	52

Total params: 28,977 (113.19 KB)

Trainable params: 28,977 (113.19 KB)

Non-trainable params: 0 (0.00 B)

[40]:
```
model_lab.compile(optimizer="adam", loss="mse", metrics=["mae"])
```

[41]:
```
history_lab = model_lab.fit(X_train, y_train, batch_size=50, epochs=epochs_lab, validation_split=0.2, verbose=1)
```

```
Epoch 1/4
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 5s 1ms/step - loss: 0.0022 - mae: 0.0261 - val_loss: 0.0011 - val_mae: 0.0227
Epoch 2/4
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 4s 1ms/step - loss: 7.4831e-04 - mae: 0.0186 - val_loss: 9.0352e-04 - val_mae: 0.0199
Epoch 3/4
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 4s 1ms/step - loss: 7.2002e-04 - mae: 0.0181 - val_loss: 8.6218e-04 - val_mae: 0.0194
Epoch 4/4
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 4s 1ms/step - loss: 7.0874e-04 - mae: 0.0179 - val_loss: 8.9643e-04 - val_mae: 0.0202
```

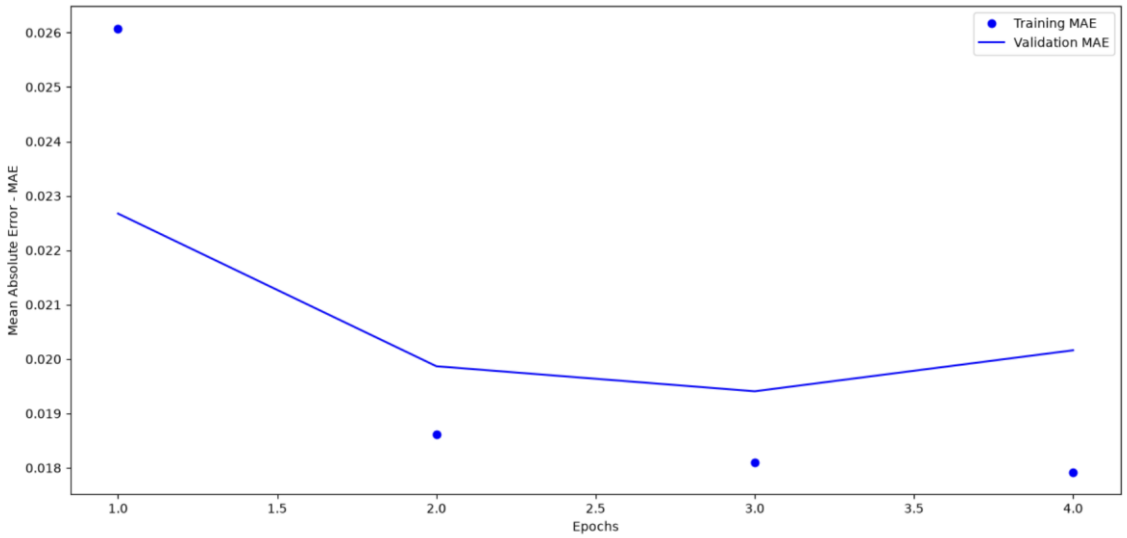
[42]:
```
mse_lab, mae_lab = model_lab.evaluate(X_test, y_test, verbose=1)
print("Lab CNN test MAE:        %.5f" % mae_lab)
print("Practical CNN test MAE:  %.5f" % mae_practical)
```

```
936/936 ━━━━━━━━━━━━━━━━━━━━ 0s 324us/step - loss: 6.5530e-04 - mae: 0.0178
Lab CNN test MAE:        0.01784
Practical CNN test MAE:  0.01639
```

[43]:
```
history_dict = history_lab.history

mean_absolute_error_values = history_dict['mae']
val_mean_absolute_error_values = history_dict['val_mae']

epochs_range = range(1, len(mean_absolute_error_values) + 1)
plt.figure(num=1, figsize=(15,7))
plt.plot(epochs_range, mean_absolute_error_values, 'bo', label='Training MAE')
plt.plot(epochs_range, val_mean_absolute_error_values, 'b', label='Validation MAE')
plt.xlabel('Epochs')
plt.ylabel('Mean Absolute Error - MAE')
plt.legend()
plt.show()
```



Lab 6

Lab 7

Lab 8

Lab 9

Lab 10

Lab 11

Lab 12